Requirements Analysis

Project: Smart E-Complaint & Public Service Management System

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Requirements Analysis

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Functional Requirements

1. User Registration and Secure Login

1.1 Citizens can create accounts using name, phone number, email, and national ID.
1.2 The system must implement secure authentication password encryption.

1.3 Only authorized users can log in and submit complaints, ensuring personal data protection.

2. Submit Complaints with Category, Description, Attachments

2.1 Citizens can select relevant service categories (e.g., roads, electricity, water).
2.2 The complaint form must allow detailed descriptions of the issue.

3. Real-Time Complaint Tracking

3.1 The system generates a unique complaint ID after submission.
3.2 Citizens can track status updates such as “Pending,” “In Progress,” or “Resolved.”

4. Post-Resolution Feedback Submission

4.1 After resolution, citizens are invited to provide feedback.
4.2 Feedback may include satisfaction ratings and written reviews.
4.3 Citizens can request to reopen the complaint if unsatisfied.

5. View Assigned Complaints

5.1 Government officers can view complaints assigned to their department or role.
5.2 Complaints can be filtered by date, priority, category, or location.
5.3 Dashboards display workload and pending responsibilities.

6. Update Complaint Status

6.1 Officers can update the complaint status as work progresses.
6.2 Status changes are instantly reflected in the citizen’s tracking interface.
6.3 This ensures transparency and keeps citizens informed.

7. Prioritize Urgent Complaints

7.1 Officers can mark complaints as high priority based on severity or risk.
7.2 Urgent issues (e.g., water leakage, electrical hazards) are fast-tracked.
7.3 Priority tagging ensures critical problems receive immediate attention.

8. Add Resolution Details

8.1 Officers must submit a resolution report after resolving the issue.
8.2 The report includes actions taken, materials used, and completion date.

9. Manage Users

9.1 Admins can create, edit, suspend, or delete user accounts.
9.2 Role-based access control must be enforced for citizens and officers.
9.3 This ensures system security and proper user management.

10. Manage Departments

10.1 Admins can create and manage government departments.
10.2 Each complaint category is mapped to a responsible department.
10.3 This ensures automatic and accurate complaint routing.

11. Assign Complaints

11.1 Admins can manually assign or reassign complaints.
11.2 This is useful when auto-routing fails or workload balancing is needed.
11.3 Proper assignment helps speed up resolution time.

12. Generate Statistical Reports

12.1 The system generates reports on received, resolved, and pending complaints.
12.2 Department performance can be analyzed.

Non-functional Requirements

Security

- Secure authentication
- Data protection
- Role-based access control

Performance

- Instant submission processing
- Real-time updates
- Multi-user handling capability

Usability

● User-friendly interface

● Easy navigation

● Minimal training required

Availability

● 24/7 web access

● Desktop & mobile support

Reliability

● Accurate record maintenance

● Regular data backups

Maintainability

● Modular system design

● Easy upgrades and feature expansion

Political

● Follow government rules and policies

● Transparency in complaint handling to reduce corruption

Cultural

• Simple design for all types of users

• Respectful and formal communication tone